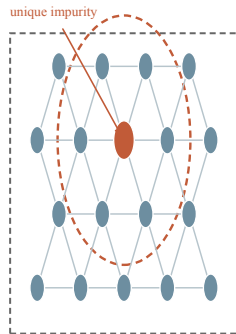


(a) Periodic defect graph

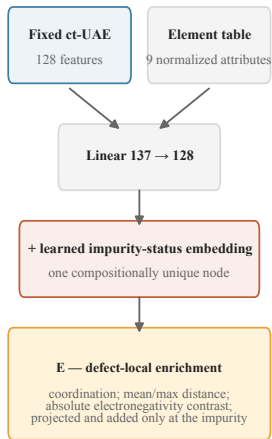


5 Å periodic pair/angle graph

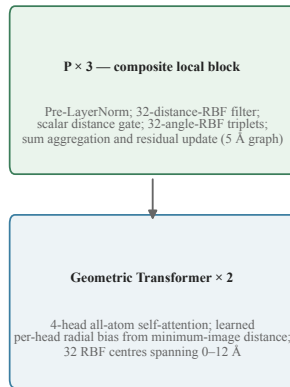
all-pairs minimum-image distances

schematic; not a selected material

(b) Atom encoding and E



(c) Local and global context



12 Å is the radial grid endpoint, not an attention cutoff

(d) G readout and prediction

